12. Merge Sort

Problem Statement

Write Recursive Approach for Merge Sort.

The problem is to implement the Merge Sort algorithm using recursion. Merge Sort is an efficient sorting algorithm that follows the divide-and-conquer approach. It divides the input array into two halves, recursively sorts each half, and then merges the sorted halves to obtain the final sorted array.

Examples

#	Input Array	Output	Explanation
1	[5, 2, 8, 3, 1, 6]	[1, 2, 3, 5, 6, 8]	The input array is divided into two halves: [5, 2, 8] and [3, 1, 6]. Each half is recursively sorted. Then, the sorted halves are merged to obtain the final sorted array.
2	[9, 4, 7, 2, 1]	[1, 2, 4, 7, 9]	The input array is divided into two halves: [9, 4] and [7, 2, 1]. Each half is recursively sorted. Then, the sorted halves are merged to obtain the final sorted array.
3	[3, 1, 2, 5, 4]	[1, 2, 3, 4, 5]	The input array is divided into two halves: [3, 1, 2] and [5, 4]. Each half is recursively sorted. Then, the sorted halves are merged to obtain the final sorted array.

Try it yourself

Try solving this question here:

Java

```java
import java.util.Arrays;

public class Solution {
    public static int[] mergeSort(int[] arr) {
      // TODO: Write your code here
      return arr;
    }
}

```

Test Case | Run Code Result | Stdout

[5,2,8,3,1,6]

Use Example Testcases | Run | Submit